Project Proposal

Project: Evaluation of Speech Enhancement Algorithms Under Different Background Noise Conditions

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1 Project Overview

Background noise can reduce the quality and intelligibility of speech recordings in environments such as cafes, traffic and other locations where the noise changes over time. Speech enhancement techniques can be used to reduce unwanted background noise while maintaining the important components of the speech signal.

This project will investigate how different speech enhancement algorithms perform under different background noise conditions. Wiener filtering and spectral subtraction will be applied to speech signals containing different types and levels of noise. The performance of each method will then be evaluated to determine how the type and severity of background noise affects the quality of the enhanced speech.

2 Background and Motivation

Speech enhancement is used to improve speech signals that have been affected by unwanted background noise. This is important in applications such as communication systems, hearing devices and speech recognition, where environmental noise can make speech more difficult to understand.

A number of signal processing techniques have been developed for speech enhancement. Spectral subtraction is a well established method which estimates the background noise spectrum and subtracts it from the noisy speech spectrum [1]. Wiener filtering is another commonly used approach which attempts to estimate the clean speech signal while reducing the effect of unwanted noise [2].

The performance of traditional speech enhancement techniques can depend on the characteristics of the noise. This can become more difficult when the background noise changes over time or contains frequency components similar to the speech signal [4]. For example relatively constant noise such as a fan may behave differently from traffic or cafe noise which can change significantly over time.

This project was chosen to investigate this behaviour under different noise conditions. Rather than testing the algorithms using only one type of noise the project will compare their performance using several environmental noise types and noise levels. This will allow the strengths and limitations of each method to be investigated under different conditions.

3 Proposed Methodology

The project will be implemented using MATLAB and will compare two speech enhancement techniques: spectral subtraction and Wiener filtering. Clean speech recordings will first be obtained and used as the reference signals for the experiment. Different background noise recordings will then be added to the clean speech to create noisy speech signals under controlled conditions.

Several types of background noise will be considered, including white noise, traffic noise and cafe noise. These noise types have different characteristics and will allow the algorithms to be tested under both relatively stationary and changing noise conditions.

The noise will also be added at different signal-to-noise ratio (SNR) levels. For example SNR levels such as -5 dB, 0 dB, 5 dB and 10 dB may be considered. This will allow the effect of increasing noise severity on the performance of each algorithm to be investigated.

The noisy speech signals will then be processed using spectral subtraction and Wiener filtering. Short time Fourier transform (STFT) analysis will be used to observe how the frequency content of the speech changes over time. Spectrograms of the clean, noisy and enhanced speech signals will also be produced to visually compare the effect of each enhancement technique.

The enhanced signals will be compared with the original clean speech using SNR improvement and spectrogram analysis. If applicable an additional speech quality measure such as the Perceptual Evaluation of Speech Quality (PESQ) may also be used. Similar objective and visual evaluation methods have previously been used to compare speech enhancement algorithms under different noise conditions [4].

The results will then be compared to determine how the performance of each algorithm changes depending on the type and severity of the background noise.

4 Expected Outcomes

The project is expected to produce a MATLAB based simulation that demonstrates the effect of spectral subtraction and Wiener filtering on speech affected by different types and levels of background noise.

The performance of each algorithm will be evaluated using measures such as SNR improvement and spectrogram analysis. The project is expected to show that the effectiveness of the algorithms may change depending on the characteristics and severity of the background noise.

The results should identify conditions where each method performs effectively as well as conditions where its performance becomes limited. This will provide a better understanding of the advantages and disadvantages of the two speech enhancement techniques when they are used in different acoustic environments.

The MATLAB implementation, results, figures and relevant project documentation will also be maintained through the GitHub repository. The final repository will provide the code required to reproduce the main results of the project.

5 Project Timeline

Table 1: Proposed project timeline.

Weeks	**Planned Work**
1–3	Select the project topic and define the main research question.
4–5	Review relevant literature and collect suitable speech and background noise recordings.
6–7	Implement spectral subtraction and Wiener filtering in MATLAB.
8–9	Test both algorithms using different noise types and SNR levels.
10–11	Evaluate and compare the results using SNR improvement, spectrograms and other suitable performance measures.
12–13	Complete the final report, organise the results and finalise the GitHub documentation.

6 References

References

[1] S. F. Boll, "Suppression of acoustic noise in speech using spectral subtraction," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 27, no. 2, pp. 113–120, 1979.

[2] P. Scalart and J. V. Filho, "Speech enhancement based on a priori signal to noise estimation," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, vol. 2, pp. 629–632, 1996.

[3] M. Berouti, R. Schwartz and J. Makhoul, "Enhancement of speech corrupted by acoustic noise," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 208–211, 1979.

[4] N. Upadhyay and A. Karmakar, "Speech enhancement using spectral subtraction-type algorithms: A comparison and simulation study," *Procedia Computer Science*, vol. 54, pp. 574–584, 2015.